

Ex. No. : 02

Date:

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A PYTHON PROGRAM TO IMPLEMENT IMPLEMENT SIMPLE LINEAR REGRESSION USING LEAST SQUARE METHOD

Aim:

A python program to implement Simple linear regression using Least Square Method

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load your dataset data =
pd.read_csv((r"C:\Users\AI_LAB\Downloads\archive (4)\Iris.csv"))

# Select two columns for Simple Linear Regression
x = data["SepalLengthCm"].values
y = data["PetalLengthCm"].values

n = len(x)
```

```

# Calculate required sums
sum_x = sum(x) sum_y = sum(y) sum_xy
= sum(x[i] * y[i] for i in range(n)) sum_x2
= sum(x[i] ** 2 for i in range(n))

# Least Squares Formula  $b1 = (n * sum\_xy - sum\_x * sum\_y) / (n * sum\_x2 - sum\_x ** 2)$ 
b1 = (sum_y - b1 * sum_x) / n

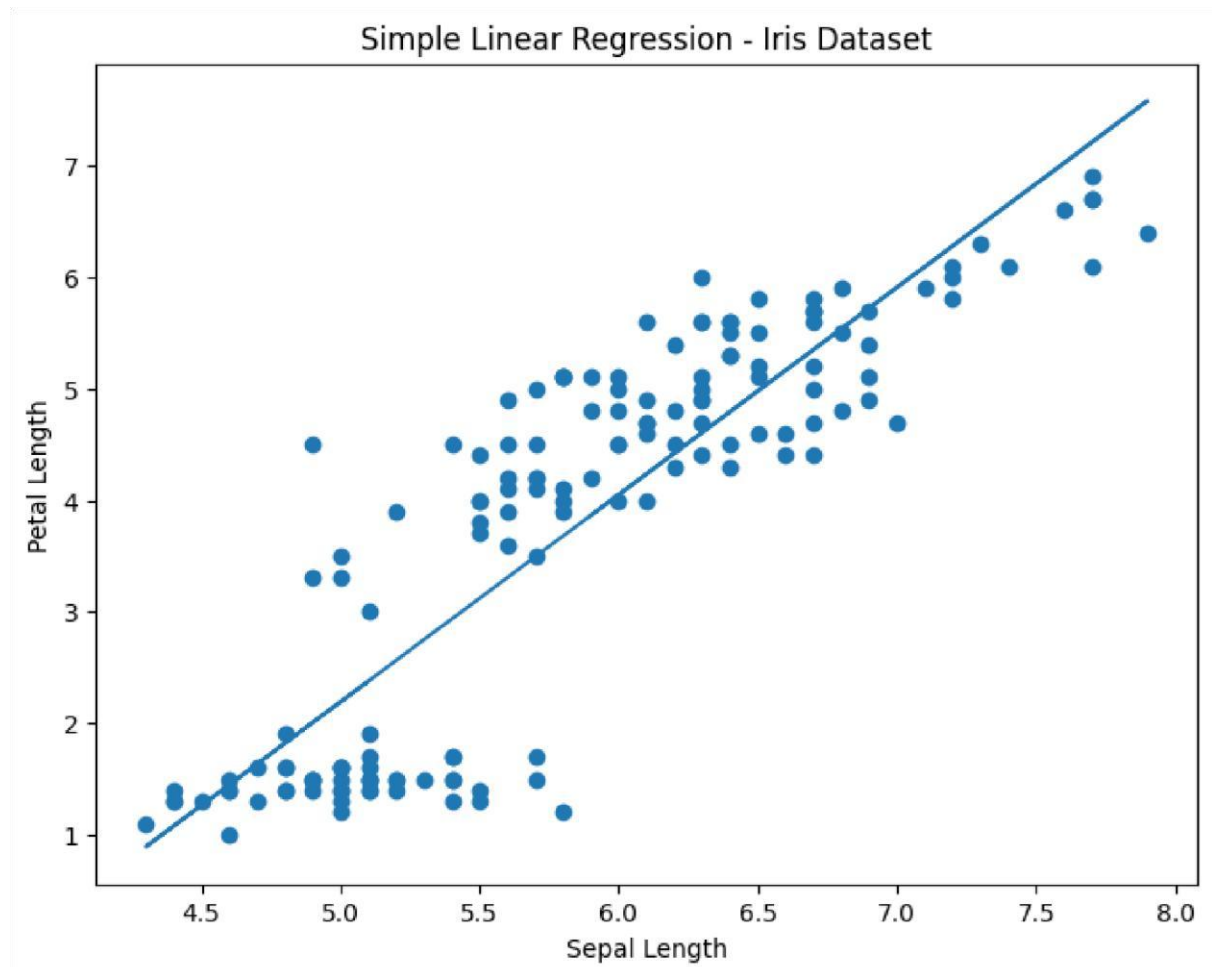
print("Slope (b1):", b1) print("Intercept
(b0):", b0)

print("\nRegression Equation:") print(f'y
= {b0:.4f} + {b1:.4f}x')

# Plotting plt.scatter(x, y) y_pred = [b0 + b1 * xi for xi in x]
plt.plot(x, y_pred) plt.xlabel("SepalLengthCm")
plt.ylabel("PetalLengthCm") plt.title("Simple Linear
Regression (Least Squares Method)") plt.show()

```

OUTPUT:



Result:

The Simple Linear Regression model was successfully implemented on the Iris dataset using the Least Square Method. The graph shows a strong linear relationship between sepal length and petal length.